



Chapter Notes: Light: Mirrors and Lenses

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Have you ever wondered why the warning "Objects in mirror are closer than they appear" is written on the side-view mirrors of cars? Or why reading glasses sometimes have a curved line on their lenses?

Let us explore these questions with Meena. On a sunny afternoon during her summer holidays, Meena visited a science centre. Among the displays, a row of curved mirrors caught her eye. To her surprise, when she looked into one, her face seemed comically large, while her brother, only a little farther away, looked upside down. At another mirror she saw a tiny version of herself staring back.



Mirrors

Why do mirrors behave this way? Through the study of spherical mirrors and lenses we will discover how light makes images appear bigger, smaller or inverted, and learn the rules that govern reflection and refraction.

What are Spherical Mirrors?

Spoon as a Mirror - a Simple Observation

A shiny metallic spoon can act like a mirror. You can see your face in it if you hold it close.



(a) (b)
Image formed in a shiny metallic spoon from its (a) Inner curved surface; (b) Outer curved surface

If you look at the **inner (curved inward) side** of the spoon, your image may appear **inverted** (upside down).

If you look at the **outer (bulging outward) side** the image appears **erect** (upright) but much **smaller** than your real face.

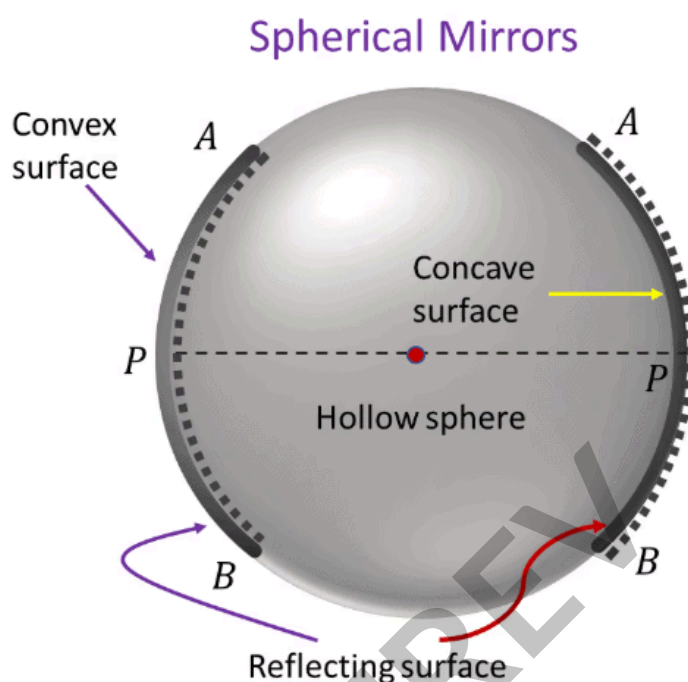
The two sides of the spoon curve differently; each side behaves like a different type of curved mirror.

Curved Mirrors and Spherical Mirrors

Mirrors shaped like parts of spheres are called **spherical mirrors**.

They are commonly used in science and everyday objects because their simple shape gives predictable image behaviour.

Definition: Spherical mirrors are mirrors whose reflecting surfaces form part of the surface of an imaginary hollow glass sphere. The reflecting surface may curve **inwards** or **outwards**, producing different mirror types.



Types of Spherical Mirrors

1. Concave Mirror

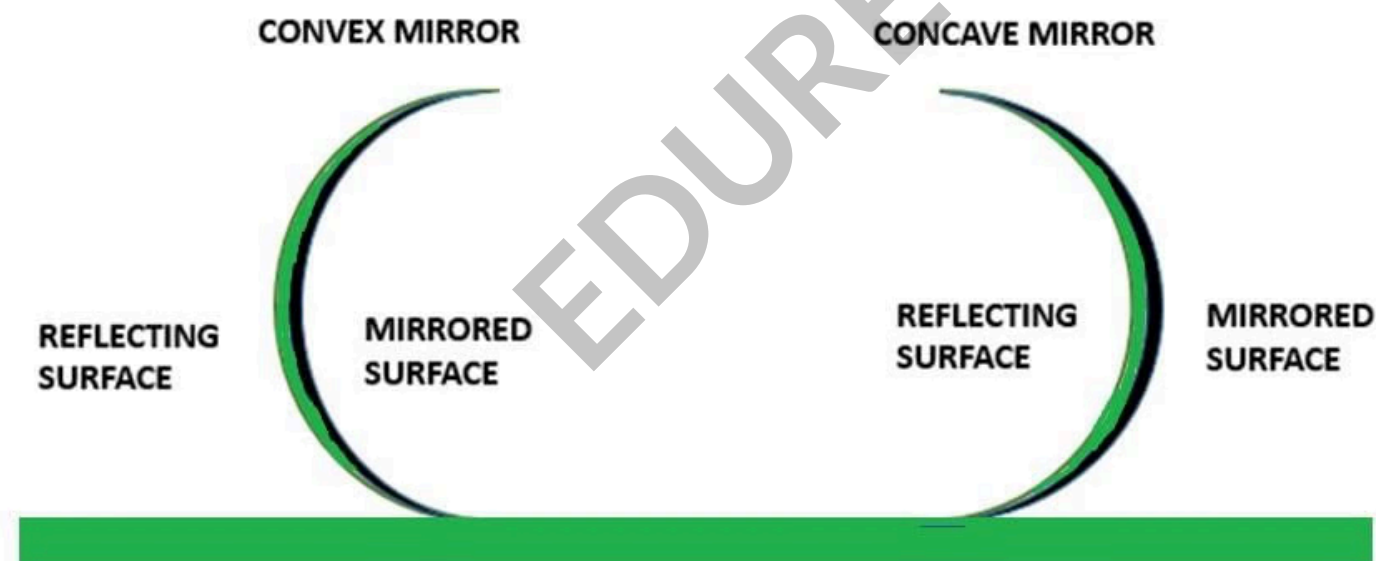
A mirror whose reflecting surface is curved **inwards**, like if you look at the inner side of the spoon, the image may appear erect and enlarged when close, but inverted when moved farther away.

It looks as if a part of a hollow sphere has been cut so the centre of the sphere lies in front of the mirror surface.

2. Convex Mirror

A mirror whose reflecting surface curves **outwards**, like the outer side of a spoon or the back of a bowl.

It behaves as if the reflecting surface is part of the outer surface of a sphere; the centre of the sphere lies behind the mirror surface.



TYPES OF SPHERICAL MIRRORS

How Spherical Mirrors Are Made

Shape: Spherical mirrors have the shape of a part of an imaginary hollow sphere.

Manufacture: They are made by grinding and polishing a flat piece of glass into a curved surface and then applying a thin **reflective coating** (for example, aluminium) on the required side.

Reflective coating placement: If the coating is applied on the outer curved surface the mirror behaves as a **concave mirror**. If the coating is on the inner curved surface the mirror behaves as a **convex mirror**.

Characteristics of Images Formed by Spherical Mirrors

Image Behaviour - Concave Mirror

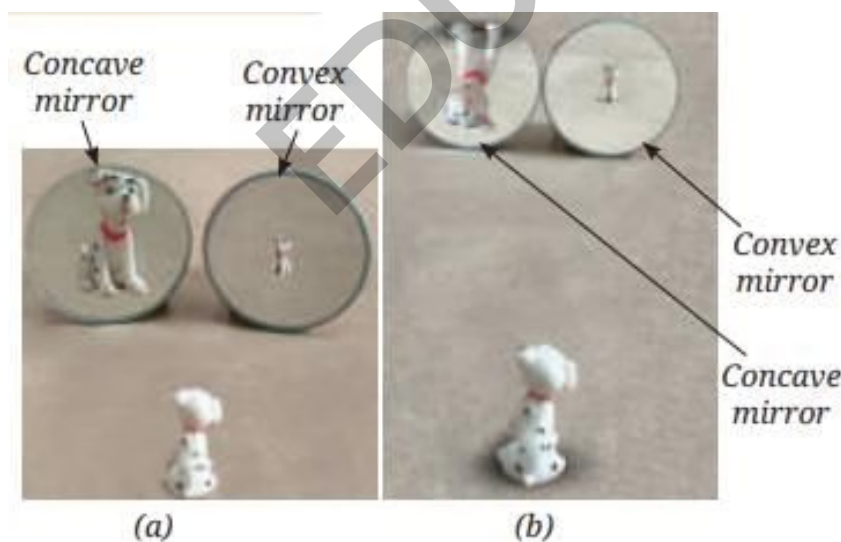
When an object is very close to a concave mirror (near the mirror), the image is **erect** and **enlarged**.

When the object is moved away from the mirror, the image becomes **inverted**. The image size changes with object distance - it may be enlarged, equal to, or smaller than the object depending on how far the object is from the mirror.

Image Behaviour - Convex Mirror

For a convex mirror at any object distance, the image is always **erect** and **diminished** (smaller).

As the object moves farther, the image size decreases slightly but remains upright and reduced.



An object placed in front of concave and convex mirrors at (a) Small distance; (b) Large distance

Common feature: Images in mirrors show **lateral inversion** (left-right reversal).

Comparison with Plane Mirrors

A **plane mirror** always forms an **erect** image the **same size** as the object and at the same distance behind the mirror as the object is in front.

Spherical mirrors (concave or convex) produce image sizes and orientations that change with object distance; concave mirrors can produce inverted images while plane mirrors cannot.

How to Identify Mirror Type by Looking at Images

Concave: If a mirror gives an enlarged upright image when you are close and an inverted image when you move away, it is a concave mirror.

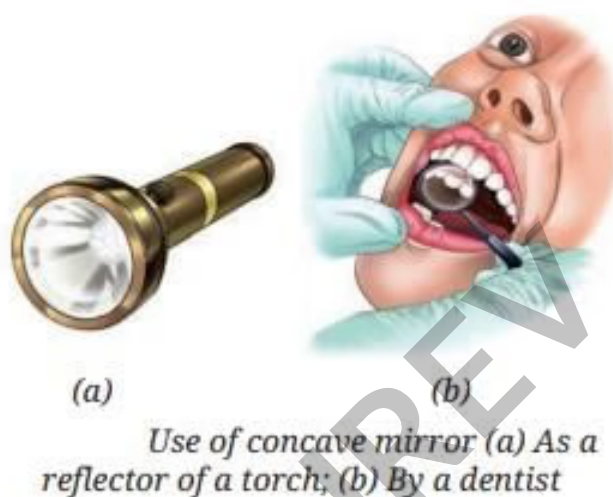
Convex: If the mirror always gives an upright small image, it is a convex mirror.

Plane: If the image is always upright and the same size as the object, it is a plane mirror.

Real-Life Uses of Spherical Mirrors

Both concave and convex mirrors are used widely because of their image-forming properties.

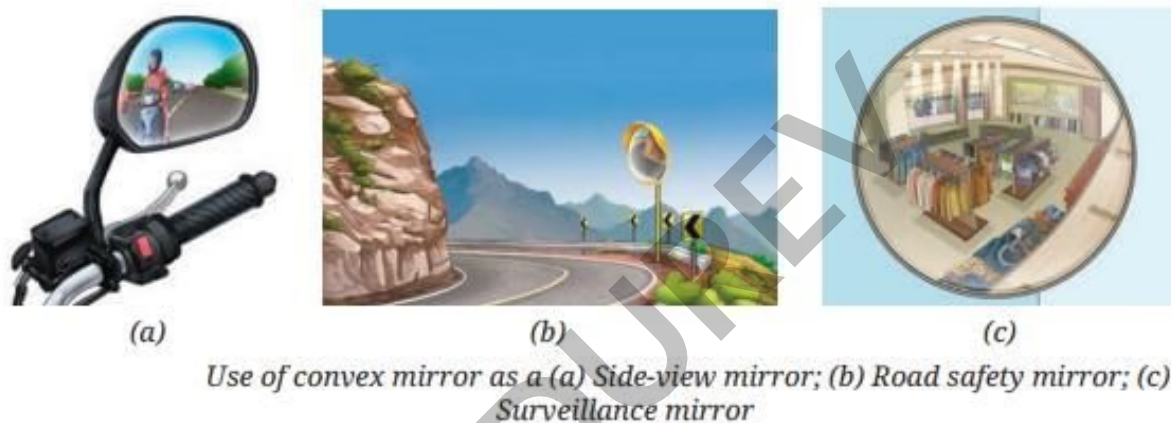
Concave Mirrors



Used as reflectors in torches, car headlights and searchlights because they **concentrate light** into a beam.

Used by dentists as dental mirrors - when held close they give an **enlarged view** of the teeth and inside of the mouth.

Convex Mirrors



Used as **side-view mirrors** on vehicles because they give a **wider field of view**, allowing drivers to see more area behind and to the side.

Placed at road intersections and sharp bends to give a view of oncoming traffic and prevent collisions.

Used in large shops and stores to monitor wide areas and deter theft.

Telescopes

Many modern telescopes are **reflecting telescopes** that use a large concave mirror as the main light collector.

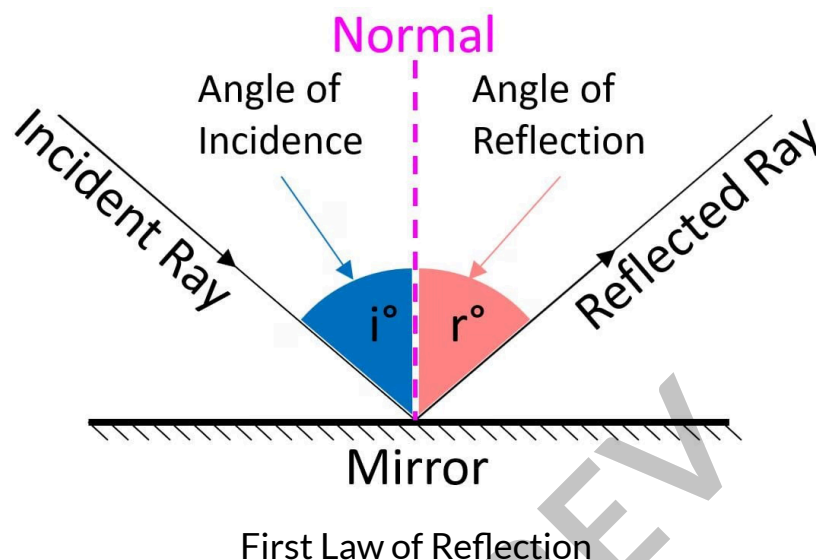
The Laws of Reflection

Reflection is the bouncing back of light from a surface. The behaviour of reflected light is described by two simple laws that apply to plane and curved mirrors.

The Two Laws of Reflection

First Law: Angle of Incidence Equals Angle of Reflection

Definition: The angle at which the incoming light ray strikes the surface (angle of incidence, **i**) is equal to the angle at which it leaves (angle of reflection, **r**). In symbols: **$i = r$** .



Key Terms

Incident ray: the incoming ray that strikes the mirror.

Reflected ray: the ray that bounces off the mirror.

Normal: an imaginary line perpendicular (at 90°) to the mirror at the point where the ray strikes.

Angle of incidence (i): the angle between the incident ray and the normal.

Angle of reflection (r): the angle between the reflected ray and the normal.

Light is represented by straight **rays** because it travels in straight lines in a uniform medium.

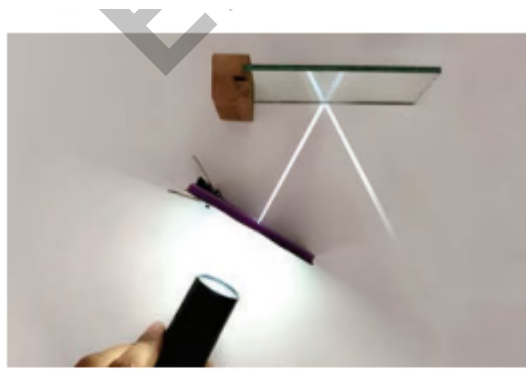
How to Verify the First Law - Simple Classroom Activity

Place a plane mirror upright on a sheet of white paper and draw its position.

Make a thin beam of light using a torch and a comb or use a narrow slit so you have a single ray that falls on the mirror.

At the point of incidence, draw the normal and measure the angle between the incident ray and the normal (i) and between the reflected ray and the normal (r).

Repeat for different incident angles and you will find $i = r$ in each case.

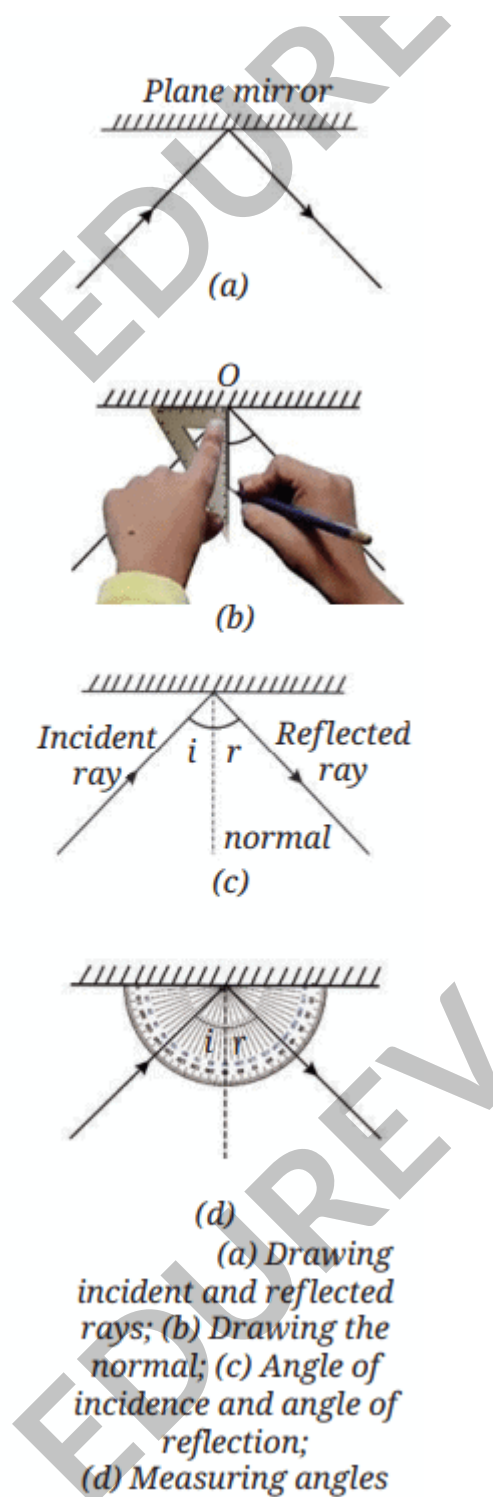


(a)



(b)

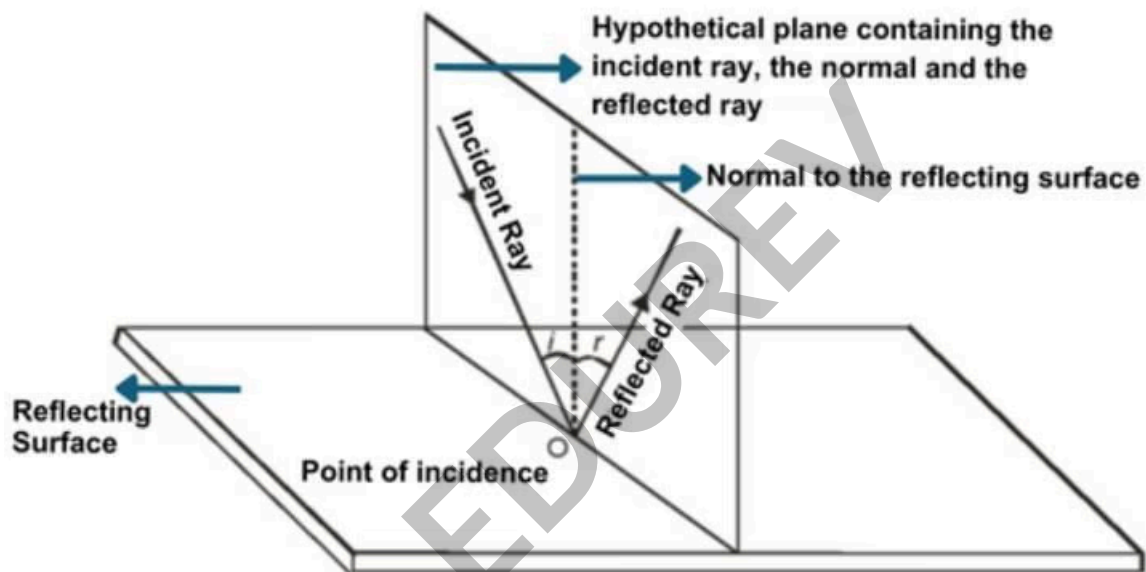
*A beam of light falling on a mirror (a) At one angle;
(b) At another angle*



Special case: If the incident ray strikes along the normal, both i and r are zero and the ray retraces its path.

Second Law: Incident Ray, Normal and Reflected Ray Lie in the Same Plane

Definition: The incident ray, the normal at the point of incidence and the reflected ray all lie in the same geometric plane.



Classroom demonstration idea

Use a plane mirror on a sheet of paper and shine a narrow beam so the path of the incident and reflected rays is visible on the paper.

If you bend the paper out of the original plane, the reflected beam disappears from the paper surface because the rays no longer lie in the same plane - this shows the importance of the plane in the second law.



(a)



(b)

(a) Reflected beam visible along the extended portion of paper; (b) Reflected beam not visible along the bent portion of paper

Applying the Laws to Spherical Mirrors

The same two laws of reflection are true for every point on a spherical mirror. Each incident ray obeys $i = r$ and lies in the same plane with the corresponding reflected ray.



(a)



(b)



(c)



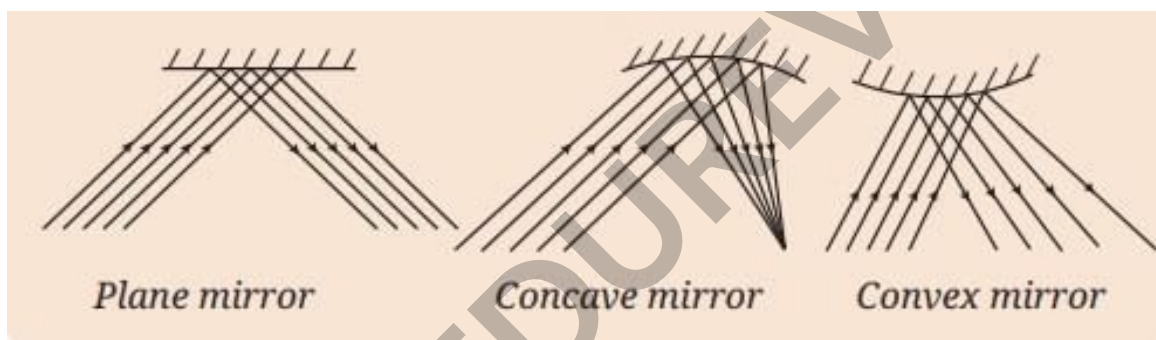
(d)

(a) Multiple slits;
Multiple parallel beams of light
fall upon — (b) Plane mirror;
(c) Concave mirror;
(d) Convex mirror

When parallel rays of light fall on a **plane mirror**, they reflect as parallel rays.

When parallel rays fall on a **concave mirror**, the reflected rays are brought together - they **converge**.

When parallel rays fall on a **convex mirror**, the reflected rays appear to come from a point behind the mirror - they **diverge**.

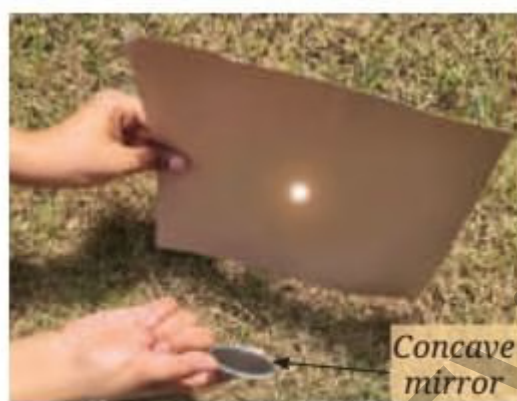


These behaviours explain how concave mirrors can focus light and convex mirrors provide a wider field of view.

Concentrating Sunlight with a Concave Mirror

Never look at the Sun or at sunlight reflected by a mirror - it can damage your eyes.

Using a concave mirror to focus sunlight on a small piece of paper will produce a sharp bright spot that can heat and eventually burn the paper after some time.



Converging sunlight on paper using a concave mirror

This demonstrates that a concave mirror can concentrate energy from parallel rays into a small area; the principle is used in solar concentrators and solar furnaces.

What is a Lens?

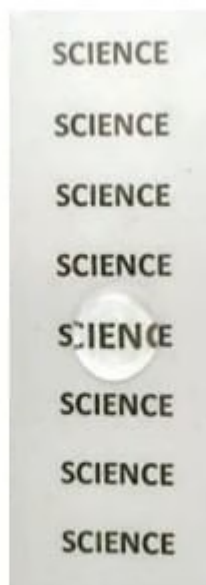
Looking through a flat glass pane does not change the apparent size of objects, but a curved transparent surface can change the way objects appear. Such curved transparent pieces are called **lenses**.

How a Water Drop Acts Like a Lens - Classroom Activity

Materials: a flat strip of glass or clear plastic, a dropper, water and a printed page.

Place a small water drop on the glass strip and hold it over printed text.

Observe the letters through the drop; they often appear **larger** because the curved water surface refracts light and acts as a simple lens.



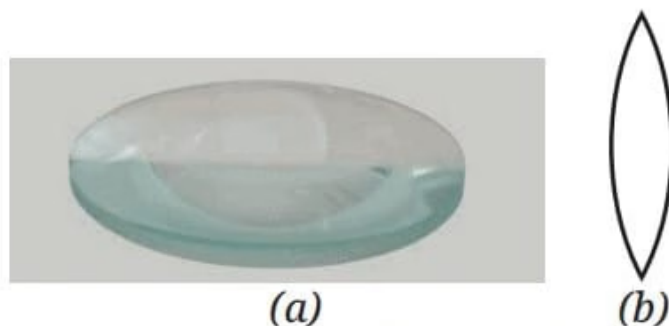
*View of text
beneath the water drop*

Definition and Types of Lenses

A **lens** is a transparent piece of glass or plastic with one or two curved surfaces that refract light.

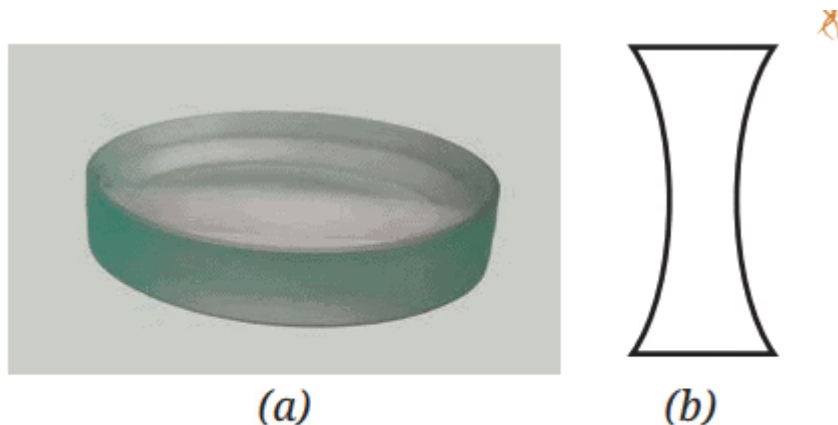


Convex lens (converging lens): thicker at the centre than at the edges.



(a) A convex lens;
(b) Its representation

Concave lens (diverging lens): thinner at the centre and thicker at the edges.



(a) A concave lens;
(b) Its representation

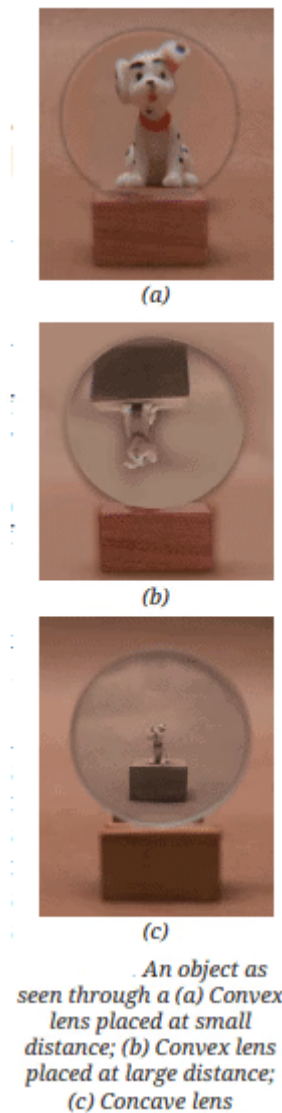
Unlike mirrors, lenses allow light to pass through; we see objects **through** a lens, not by reflection from its surface.

How Objects Look Through Lenses - Observations

Using a convex and a concave lens in a simple setup shows how image size and orientation change with object distance.

Convex lens: When the object is close to the lens it may form an **erect enlarged** image (used as a magnifying glass). When the object is farther away the image can become **inverted** and may be enlarged or diminished depending on the distance.

Concave lens: For any object distance, a concave lens forms an **erect diminished** image (smaller than the object).

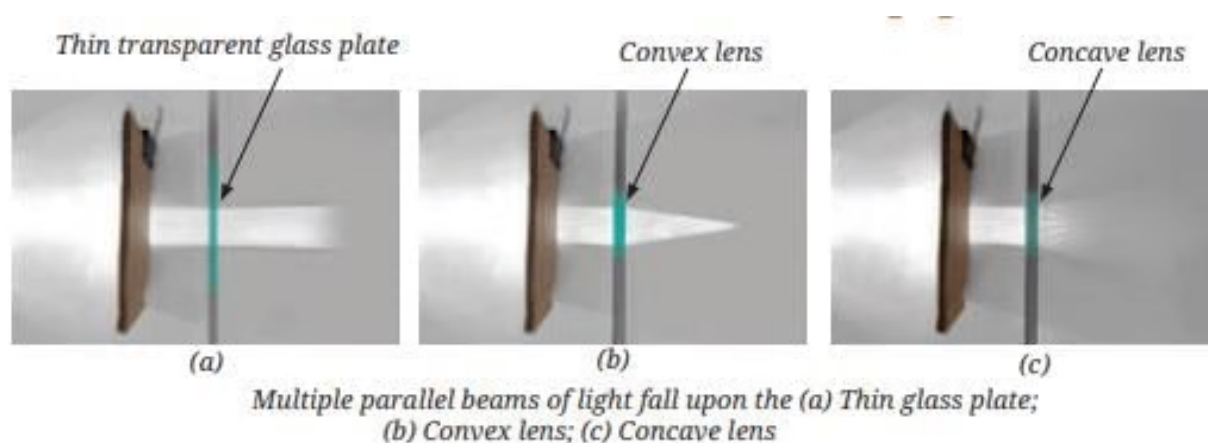


Do Lenses Converge or Diverge Light?

A thin transparent plate does not change the direction of parallel beams - they remain parallel.

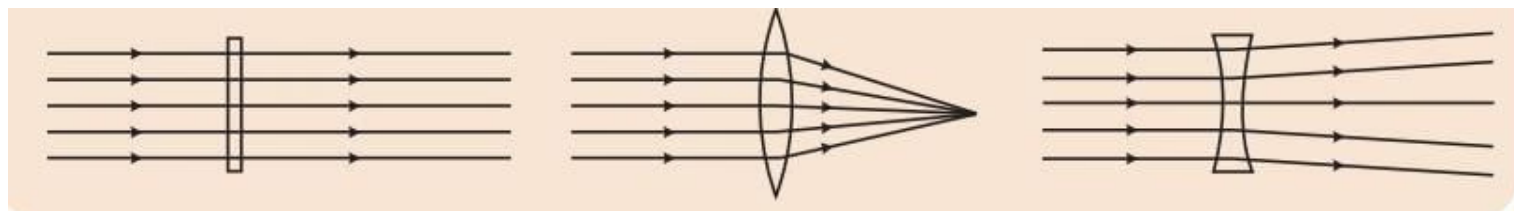
A **convex (converging) lens** makes parallel rays come together at a point (focus).

A **concave (diverging) lens** makes parallel rays spread out as if they came from a point.



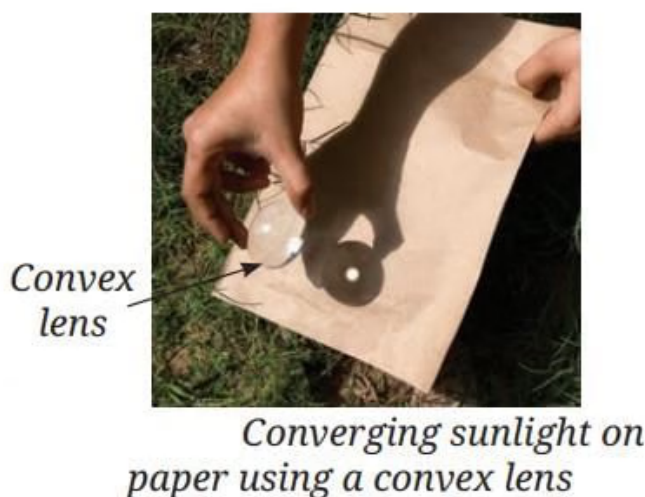
Drawing Ray Diagrams for Lenses

Simple ray diagrams show how rays behave: through a thin glass plate rays remain straight, through a convex lens they converge, and through a concave lens they diverge. Ray diagrams help predict image position, size and orientation for lenses in the same way as for mirrors.



Can a Convex Lens Burn Paper?

Yes. Like a concave mirror, a convex lens can concentrate sunlight into a small hot spot and can burn paper when sunlight is focused correctly.



This is the same principle used in burning-glass demonstrations and in focusing sunlight for solar heating.

Real-Life Uses of Lenses

Eyeglasses: Convex lenses help long-sighted (hypermetropic) eyes, while concave lenses help short-sighted (myopic) eyes.

Cameras: Use lenses to form sharp images on film or sensors.

Telescopes and microscopes: Use combinations of lenses (and sometimes mirrors) to magnify distant or tiny objects.

The human eye: contains a natural convex lens that changes shape to focus on near and far objects.

Key Points to Remember

Concave mirror images: Can be enlarged, diminished or same size; can be erect or inverted depending on object distance.

Convex mirror images: Always erect and diminished regardless of object distance.

Laws of reflection: (1) Angle of incidence equals angle of reflection ($i = r$). (2) Incident ray, normal, and reflected ray lie in the same plane.

The laws of reflection are valid for plane, concave and convex mirrors.

Light behaviour with mirrors: Concave mirrors **converge** rays (focus light); convex mirrors **diverge** rays (wider view).

Convex lens images: Can be enlarged, diminished or same size; image can be erect or inverted depending on object distance.

Concave lens images: Always erect and diminished regardless of object distance.

Behaviour of lenses: Convex lenses **converge** rays (used in magnifying glasses, cameras, eyes); concave lenses **diverge** rays (used in some glasses and optical devices).